# Basic SRS / Implementation Plan

Version 1.0 • Agriculture case

## Purpose and boundary

Version 1 specification for a hackathon prototype supporting one crop and 3–4 markets. The system estimates net realisation from entered/reported inputs; it does not guarantee a price, forecast a guaranteed best selling time, trade, or replace authorised advice.

## Users

• Farmer or farmer-group representative.

• FPO/field facilitator who helps verify quotes and costs.

• Prototype administrator who audits data and test outcomes.

## Inputs / outputs

• Inputs: crop, grade, quantity and unit, origin, deadline (optional), candidate market price, price type, source/time, costs and optional explicit loss assumption.

• Outputs: comparable estimated net realisation per market; arithmetic breakdown; data freshness, missing-input and source-conflict flags; verification prompt; timestamped cached view.

## Workflow

Enter lot details → retrieve or select price records → validate crop/grade/unit/time → enter or confirm costs → calculate deterministically → show comparison and uncertainty → farmer/facilitator verifies information → farmer chooses → optionally record actual outcome for evaluation.

## Functional requirements

• FR1: capture crop, grade, quantity/unit and origin.

• FR2: display market prices with source, price type, unit and observation/retrieval times when available.

• FR3: reject or flag incomparable crop/grade/unit records.

• FR4: allow editing of transport, loading, commission and other selling costs.

• FR5: convert units correctly and calculate net realisation deterministically.

• FR6: rank only when required inputs are sufficiently complete; otherwise show incomplete/scenario-based output.

• FR7: flag stale, missing and conflicting records; never invent prices or silently treat unknown costs as zero.

• FR8: explain the comparison in plain language and expose assumptions.

• FR9: show cached data offline with timestamp and non-live warning.

• FR10: optionally record actual sale results; no payment or transaction execution.

## AI versus human responsibility

• AI: optional Hindi/mixed-language input parsing and plain-language explanation grounded in displayed data.

• Code: unit conversion, formula, freshness policy, validation and ranking.

• Human: verify lot/grade, obtainable quote, costs and route practicality; farmer retains final decision.

• AI must not invent a missing price, turn an uncertain estimate into a guarantee, or independently prescribe harvest/financial action.

## Data model and constraints

Price record fields: commodity, variety/grade if available, market, price type (minimum/maximum/modal/transaction quote), value, unit, observation time, source, retrieval time, verification state and conflict state. These price types are not interchangeable. Freshness thresholds must be configurable and disclosed. Public-source access, licensing, API limits, update cadence and local coverage must be verified.

## Non-functional requirements

• Mobile-first and low typing burden; plain-language and Hindi-friendly wording.

• Traceable calculations: each result links to its inputs and formula.

• Repeatable results with unit and acceptance tests.

• Offline resilience with explicit data age.

• Data minimisation; avoid storing precise location unless necessary.

• Keep credentials out of client code and source control.

## Failure handling

• Source unavailable → use timestamped cache with non-live warning, or report no usable data.

• Missing price/unit/time → mark incomplete; no fabricated ranking.

• Conflicting sources → show both and request verification.

• Unknown cost → ask for input or show a labelled scenario; never silently use zero.

• Invalid quantity/unit → block calculation and explain correction.

• Ambiguous AI-parsed field → ask user to confirm before calculating.

## Acceptance tests

| Test | Expected result |
|---|---|
| Higher price but higher transport cost | Rank using net realisation, not headline price. |
| Stale price | Visible stale flag and verification prompt. |
| Unknown selling cost | No silent zero; request input or label scenario. |
| ■/quintal vs kg | Correct conversion using 1 quintal = 100 kg. |
| Conflicting price sources | Show conflict; do not silently choose one. |
| Offline use | Cached timestamp and non-live label remain visible. |
| Ambiguous parsed input | Require confirmation before calculation. |

## Version 1 exclusions

• Future-price prediction / automated harvest timing.

• Automated buying, selling, payments or buyer contracts.

• Nationwide coverage or claims of real-time coverage without verification.

• Automated crop diagnosis or agronomic prescriptions.

• Claims of increased income or reduced waste before evaluation.

## Implementation plan

• 1. Audit data for one crop and 3–4 markets.

• 2. Implement calculation and unit tests.

• 3. Build the input/comparison interface.

• 4. Add data-quality checks and timestamped cache.

• 5. Run acceptance tests and a supervised shadow test.

• 6. Report results, failures, limitations and next-step evidence needs.